

```
192.168.0.10 - Remote Desktop Connection
date
#check_design -checks pre_route_stage
check routability

# Run the following commands to set the options.
set_app_options -name route.detail.timing_driven -value true
set_app_options -name route.track.timing_driven -value true
set_app_options -name route.track.crosstalk_driven -value true
set_app_options -name route.global.timing_driven -value true

# Define the routing layers that needs to be used for routing
set_ignored_layers -max_routing_layer M7 -min_routing_layer M2

#Set the options to allow routing below the minimum layer only for pin connections.
set_app_options -name route.common.global_min_layer_mode -value allow_pin_connection

#Set the option to use a layer above the maximum layer as soft. This sets the tool to routing above the maximum layer is discouraged, but not disallowed.
set_app_options -name route.common.global_max_layer_mode -value soft

#Enabling the Crosstalk aware and Delay Calculations
set_app_options -name time.si_enable_analysis -value true
set_app_options -name time.enable_si_timing_windows -value true
set_app_options -name time.enable_ccs_rcv_cap -value true

#
set_attribute [get_layers M1] routing_direction vertical
create_via_def VIA22_Cu2on_short -cut_layer VIA1 -cut_size {0.03 0.03} -upper_layer M2 -lower_layer M1 -lower_enclosure {0.002 0.025} -upper_enclosure {0.025 0.002} -is_default
set_app_options -name route.common.connect_within_pins_by_layer_name -value {(M1 via_standard_cell_pins)}

#
set_attribute [get_layers M2] routing_direction horizontal
create_via_def VIA23_Cu2on_short -cut_layer VIA2 -cut_size {0.03 0.03} -upper_layer M3 -lower_layer M2 -lower_enclosure {0.002 0.025} -upper_enclosure {0.025 0.002} -is_default
set_app_options -name route.common.connect_within_pins_by_layer_name -value {(M2 via_standard_cell_pins)}

#route_global
#route_track
#route_detail
route_auto -max_detail_route_iterations 10

#route_auto
save_block -as initial_route_newlib
route_opt
save_block -as route_done_newlib

### helps to reduce drcs
route_eco -open_net_driven true -reuse_existing_global_route true -utilize_dangling_wires true

# VT distribution
#report_threshold_voltage_group
```

192.168.0.10 - Remote Desktop Connection

cts.tcl (-:/jbi_14nm/pd/scripts) - GVIM1

```
File Edit Tools Syntax Buffers Window Help
date
set_clock_uncertainty -setup 0.0909 [get_clocks cmp_gclk]
set_clock_uncertainty -hold 0.1818 [get_clocks cmp_gclk]

set_clock_uncertainty -setup 0.1111 [get_clocks jbus_gclk]
set_clock_uncertainty -hold 0.2222 [get_clocks jbus_gclk]

update_timing -full

#Before performing CTS, execute the following command and analyze the report
check_design -checks pre_clock_tree_stage
# set NDR
#create_routing_rule clk_rule -widths {M6 0.112 M7 0.112 } -spacings {M6 0.112 M7 0.112 }
create_routing_rule clk_rule -widths {M6 0.18 M7 0.18 } -spacings {M6 0.08 M7 0.08 }
#check_clock_tree

# specify clock tree cell list
#set_lib_cell_purpose -exclude cts [get_lib_cells]
set_lib_cell_purpose -include cts [get_lib_cells "saedi4rvt_tt0p6v125c/SAEDRVT14_BUF_5_2 saedi4rvt_tt0p6v125c/SAEDRVT14_BUF_5_4 saedi4rvt_tt0p6v125c/SAEDRVT14_BUF_5_6 saedi4rvt_tt0p6v125c/SAEDRVT14_BUF_5_8"]

#Specify Max fanout
set_app_options -name cts.common_max_fanout -value 30

# set clock target skew and latency
set_clock_tree_options -clocks [all_clocks] -target_latency 0.4 -target_skew 0.05
#set_clock_tree_options -clocks [get_clocks -filter "is_virtual==false"] -target_latency 0.25 -target_skew 0.03

set_clock_routing_rules -clocks [all_clocks] -net_type {internal} -rules clk_rule -min_routing_layer M6 -max_routing_layer M7
set_clock_routing_rules -clocks [all_clocks] -net_type {root} -rules clk_rule -min_routing_layer M6 -max_routing_layer M7

clock_opt

# Make the logical connection of PG nets for all the standard cells
connect_pg_net -net VDD [get_pins -hier * -filter "name == VDD"]
connect_pg_net -net VSS [get_pins -hier * -filter "name == VSS"]

report_constraints -all_violators
report_clock_tree_options
report_clock
report_clock_qor
report_qor -summary
report_timing -delay_type min
report_timing -delay_type max
date
save_block -as cts_done
~
~
```

12:22 PM 8/15/2025

192.168.0.10 - Remote Desktop Connection

data_preparation.tcl (~jbi_14nm/pd/scripts) - GVIM1

```
File Edit Tools Syntax Buffers Window Help
##### Setting Up Target & Link Libraries #####
source ../scripts/common_setup.tcl

##### Creating library of the block #####

set link_library $LINK_LIBRARY_FILES_CLG
set target_library $TARGET_LIBRARY_FILES_CLG

create_lib ../lib/jbi1 -ref_libs $NDM_REFERENCE_LIB_DIRS_CLG -technology $TECH_FILE

#read netlist. Below netlist will be available in the location, only after completing the synthesis step successfully. Else, it will error out complaining, file not found.
read_verilog ../synth/outputs/jbi.vg

#set current design -top level module name
current_design jbi

#linking library + netlist
link

#defining attributes for metal layers -- HWV (preferable) or VHV
define_user_attribute -type string -name routing_direction -classes routing_rule

#for vertical
set_attribute -objects [get_layers {M2 M4 M6 M8 MRDL}] -name routing_direction -value horizontal

#for horizontal
set_attribute -objects [get_layers {M1 M3 M5 M7 M9}] -name routing_direction -value vertical

set_wire_track_pattern -site_def unit -layer M1 -mode uniform -mask_constraint {mask_two mask_one} \
-coord 0.037 -space 0.074 -direction horizontal

#reading TLU+ file ---max file
read_parasitic_tech -tlup /home/tools/14nm/tech/star_rc/max/saed14nm_1p9m_cmax.tluplus -layermap /home/tools/14nm/tech/star_rc/saed14nm_tf_itf_tluplus.map -name worst_para

#min file
read_parasitic_tech -tlup /home/tools/14nm/tech/star_rc/min/saed14nm_1p9m_cmin.tluplus -layermap /home/tools/14nm/tech/star_rc/saed14nm_tf_itf_tluplus.map -name best_para

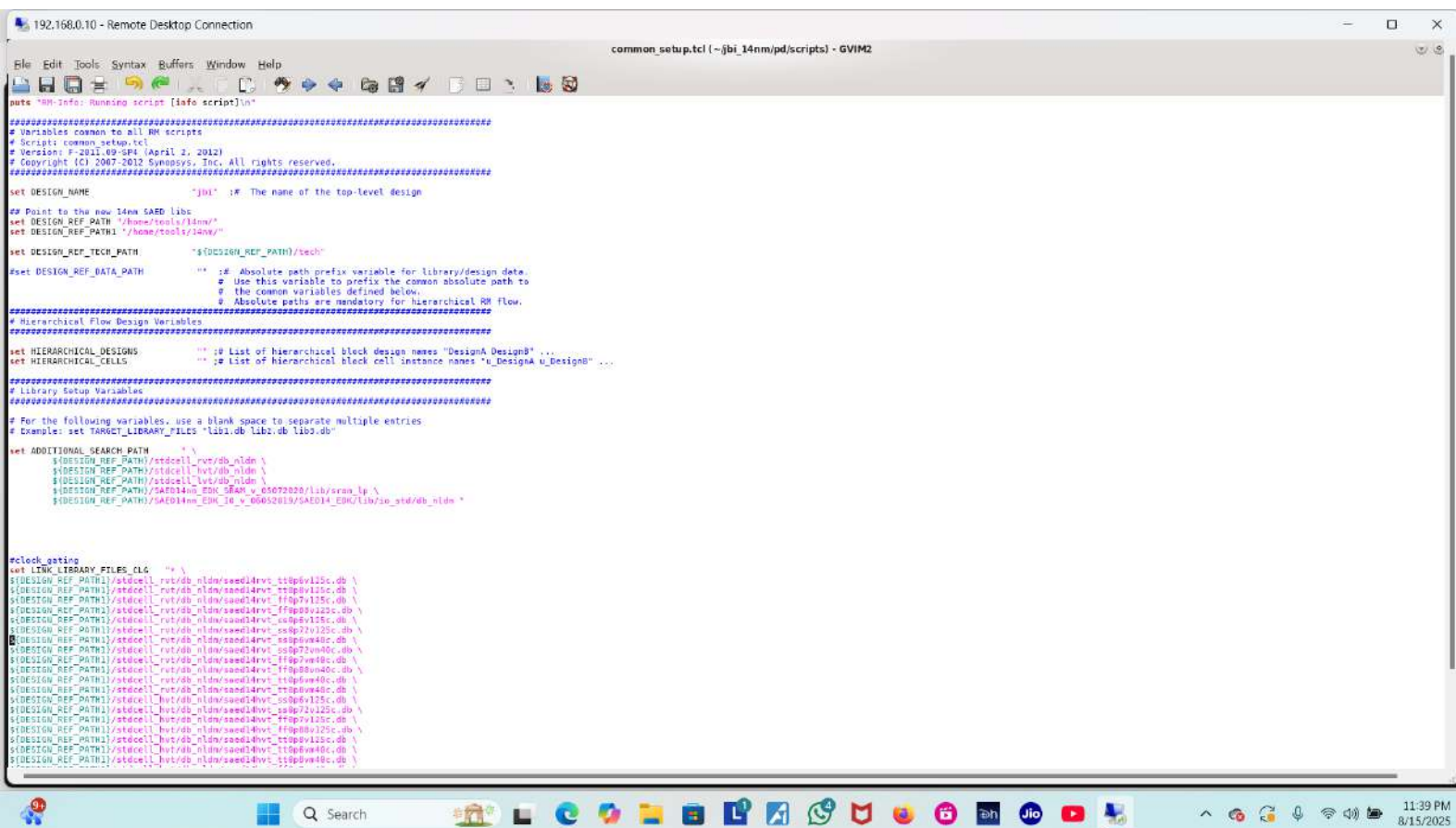
#reading the timing constraints/.sdc file from mcm(multi mode multi corner) file. The output SDC file will be available only after running synthesis step. Else SDC will not be loaded, but fail complaining file not found.
source ../scripts/mcm.tcl

#save_block -as mmsoc_data1
save_block -as import_done

save_lib

#close_blocks -force
#close_lib
```

7:44 PM
8/15/2025



```
192.168.0.10 - Remote Desktop Connection

set ADDITIONAL_LINK_LIB_FILES "
$(DESIGN_REF_PATH)/SAED14nm_EDK_SRAM_v_05072020/lib/sram/logic_synth/single/seed14sram_ff6p00v125c.db \
seed14sram_v0_tfp16v125c_2p75v.db"

set NDM_REFERENCE_LIB_DIRS_CLG "
$(DESIGN_REF_PATH)/stdcell_rvt/ndm/seed14rvt_frame_only.ndm \
$(DESIGN_REF_PATH)/stdcell_hvt/ndm/seed14hvt_frame_only.ndm \
$(DESIGN_REF_PATH)/stdcell_lvt/ndm/seed14lvt_frame_only.ndm \
$(DESIGN_REF_PATH)/stdcell_slvt/ndm/seed14slvt_frame_only.ndm \
$(DESIGN_REF_PATH)/SAED14nm_EDK_SRAM_v_05072020/lib/sram/ndm/seed14sram_inu_frame_only.ndm \
$(DESIGN_REF_PATH)/SAED14nm_EDK_SRAM_v_05072020/lib/sram/ndm/seed14sram_inu_frame_only.ndm \
$(DESIGN_REF_PATH)/SAED14nm_EDK_IO_v_06052019/SAED14_EDK/lib/so_std/ndm/seed14io_fc_frame_only.ndm \
$(DESIGN_REF_PATH)/SAED14nm_EDK_IO_v_06052019/SAED14_EDK/lib/so_std/ndm/seed14io_ub_frame_only.ndm \
$(DESIGN_REF_PATH)/SAED14nm_EDK_FLL_v_04052019/SAED14_EDK/lib/pl2/ndm/seed14pl2_frame_only.ndm \
"

set HW_REFERENCE_CONTROL_FILE "" ;# Reference Control file to define the HW ref libs

set TECH_FILE "$(DESIGN_REF_PATH)/technew_tf/seed14nm_tfp16v125c.tcl" ;# Milkyway technology file
set MAP_FILE "$(DESIGN_REF_PATH)/techstar_rc/seed14nm_tf_tfp16v125c.map" ;# Mapping file for TLuplus
set TLUPPLUS_MAX_FILE "$(DESIGN_REF_PATH)/techstar_rc/max/seed14nm_tfp16v125c.tluplus" ;# Max TLuplus file
set TLUPPLUS_MIN_FILE "$(DESIGN_REF_PATH)/techstar_rc/min/seed14nm_tfp16v125c.tluplus" ;# Min TLuplus file
set GDS_MAP_FILE "$(DESIGN_REF_PATH)/tech/milkyway/seed14nm_tfp16v125c.gds" ;# Min TLuplus file
set STD_CELL_GDS ""

set NDM_POWER_NET "VDD" ;#
set NDM_POWER_PORT "VDD01" ;#
set NDM_GROUND_NET "VSS" ;#
set NDM_GROUND_PORT "VSS01" ;#

set MIN_ROUTING_LAYER "M2" ;# Min routing layer
set MAX_ROUTING_LAYER "M7" ;# Max routing layer

#RH variable for IOC SAED library and design input data
set IIC_INPUT_DATA "global/scratch/scraper/PD_Fest_2012/initial_design/dha"

set LIBRARY_DONT_USE_FILE ".../DATA_SAED/use_lib.tcl" ;# Tcl file with library modifications for dont_use

#####
# Multi-Voltage Common Variables
#
# Define the following HW common variables for the RH scripts for multi-voltage flows.
# Use as few or as many of the following definitions as needed by your design.
#####

set PD1 "" ;# Name of power domain/voltage area 1
set PD1_CELLS "" ;# Instances to include in power domain/voltage area 1
set VAI_COORDINATES {} ;# Coordinates for voltage area 1
set NDM_POWER_NET1 "VDD01" ;# Power net for voltage area 1
set NDM_POWER_PORT1 "VDD01" ;# Power port for voltage area 1

set PD2 "" ;# Name of power domain/voltage area 2
set PD2_CELLS "" ;# Instances to include in power domain/voltage area 2
set VAI2_COORDINATES {} ;# Coordinates for voltage area 2
set NDM_POWER_NET2 "VDD02" ;# Power net for voltage area 2
set NDM_POWER_PORT2 "VDD02" ;# Power port for voltage area 2

set PD3 "" ;# Name of power domain/voltage area 3
set PD3_CELLS "" ;# Instances to include in power domain/voltage area 3

190.1 9

scripts : tclsh - Konsole [No Name] - OVIM routing.tcl (~/.jbl_14nm/jd/scripts) - GV common_setup.tcl (~/.jbl_14nm/jd/scripts) 11:39 PM 8/15/2025
```

```
192.168.0.10 - Remote Desktop Connection
set SRAM_SINGLE_GDS "${DESIGN_REF_PATH}/SAED14nm_BDK_SRAM_v_05072020/lib/sram/gds/single.gds"
set SRAMP_SINGLELP_GDS "${DESIGN_REF_PATH}/SAED14nm_BDK_SRAM_v_05072020/lib/sram/lp/gds/singlelp.gds"
set NDM_POWER_NET "VDD" ;#
set NDM_POWER_PORT "VDD" ;#
set NDM_GROUND_NET "VSS" ;#
set NDM_GROUND_PORT "VSS" ;#
set MIN_ROUTING_LAYER "M2" ;# Min routing layer
set MAX_ROUTING_LAYER "M7" ;# Max routing layer

##RH variable for ICC SAED Library and design input data
set ICC_INPUT_DATA "/global/scratch/escalver/PD_Test_2012/initial_design/dhw"

set LIBRARY_DONT_USE_FILE ".../DATA_SAED/use_tie.tcl" ;# Tcl file with library modifications for dont_use

#####
# Multi-Voltage Common Variables
#
# Define the following MV common variables for the RH scripts for multi-voltage flows.
# Use as few or as many of the following definitions as needed by your design.
#####

set PD1 "" ;# Name of power domain/voltage area 1
set PD1_CELLS "" ;# Instances to include in power domain/voltage area 1
set VA1_COORDINATES {} ;# Coordinates for voltage area 1
set NDM_POWER_NET1 "VDD01" ;# Power net for voltage area 1
set NDM_POWER_PORT1 "VDD01" ;# Power port for voltage area 1

set PD2 "" ;# Name of power domain/voltage area 2
set PD2_CELLS "" ;# Instances to include in power domain/voltage area 2
set VA2_COORDINATES {} ;# Coordinates for voltage area 2
set NDM_POWER_NET2 "VDD02" ;# Power net for voltage area 2
set NDM_POWER_PORT2 "VDD02" ;# Power port for voltage area 2

set PD3 "" ;# Name of power domain/voltage area 3
set PD3_CELLS "" ;# Instances to include in power domain/voltage area 3
set VA3_COORDINATES {} ;# Coordinates for voltage area 3
set NDM_POWER_NET3 "VDD03" ;# Power net for voltage area 3
set NDM_POWER_PORT3 "VDD03" ;# Power port for voltage area 3

set PD4 "" ;# Name of power domain/voltage area 4
set PD4_CELLS "" ;# Instances to include in power domain/voltage area 4
set VA4_COORDINATES {} ;# Coordinates for voltage area 4
set NDM_POWER_NET4 "VDD04" ;# Power net for voltage area 4
set NDM_POWER_PORT4 "VDD04" ;# Power port for voltage area 4

puts "RH-Info: Completed script [info script]\n"
```

223.1-0

scripts : tcsh - Konsole [No Name] - GVIM routing.tcl (~/.jbl_14nm/jd/scripts) - GV common_setup.tcl (~/.jbl_14nm/jd/scripts)

11:40 PM 8/15/2025

192.168.0.10 - Remote Desktop Connection

data_preparation.tcl (-jbi_14nm/pd/scripts) - GVIM3

```
File Edit Tools Syntax Buffers Window Help
##### Setting Up Target & Link Libraries #####
source ../scripts/common_setup.tcl

##### Creating library of the block #####

set link_library $LINK_LIBRARY_FILES_CLG
set target_library $TARGET_LIBRARY_FILES_CLG

create_lib ../lib/jbi1 -ref_libs $NDM_REFERENCE_LIB_DIRS_CLG -technology $TECH_FILE

#read netlist. Below netlist will be available in the location, only after completing the synthesis step successfully. Else, it will error out complaining, file not found.
read_verilog ../../synth/outputs/jbi.vg

#set current design -top level module name
current_design jbi

#linking library + netlist
link

#defining attributes for metal layers -- HWH (preferable) or VHV
define_user_attribute -type string -name routing_direction -classes routing_rule

#for vertical
set_attribute -objects [get_layers {M2 M4 M6 M8 M10}] -name routing_direction -value horizontal

#for horizontal
set_attribute -objects [get_layers {M1 M3 M5 M7 M9}] -name routing_direction -value vertical

set_wire_track_pattern -site_def unit -layer M1 -mode uniform -mask_constraint {mask_two mask_one} \
-coord 0.037 -space 0.074 -direction horizontal

#reading TLU+ file ---max file
read_parasitic_tech -tlup /home/tools/14nm/tech/star_rc/max/saed14nm_1p9n_Cmax.tluplus -layermap /home/tools/14nm/tech/star_rc/saed14nm_tf_itf_tluplus.map -name worst_para

#min file
read_parasitic_tech -tlup /home/tools/14nm/tech/star_rc/min/saed14nm_1p9n_Cmin.tluplus -layermap /home/tools/14nm/tech/star_rc/saed14nm_tf_itf_tluplus.map -name best_para

#reading the timing constraints/.sdc file from mcnmulti mode multi corner ifile. The output SDC file will be available only after running synthesis step. Else SDC will not be loaded, but fail complaining file not found.
source ../scripts/mcnm.tcl

#save_block -as minsoc_data1
save_block -as import_done

save_lib

#close_blocks -force
#close_lib
```

11:40 PM 8/15/2025

192.168.0.10 - Remote Desktop Connection

mcmmt.tcl (-jbi_14nm/pd/scripts) - GVIM4

```
File Edit Tools Syntax Buffers Window Help
set constraints_file "../../../synth/outputs/jbi.sdc"

remove_corners -all
remove_modes -all
remove_scenarios -all
remove_propagated_clocks [all_clocks]
remove_propagated_clocks [get_ports]
remove_propagated_clocks [get_pins -hierarchical]

create_corner slow
create_corner fast

read_parasitic_tech \
-tlup /home/tools/14nm/tech/star_rc/min/saed14nm_lp9n_cmin.tluplus \
-layermap /home/tools/14nm/tech/star_rc/saed14nm_tf_itf_tluplus.map \
-name tlup_min

read_parasitic_tech \
-tlup /home/tools/14nm/tech/star_rc/max/saed14nm_lp9n_cmax.tluplus \
-layermap /home/tools/14nm/tech/star_rc/saed14nm_tf_itf_tluplus.map \
-name tlup_max

set_parasitics_parameters \
-early_spec tlup_max \
-late_spec tlup_max \
-early_temperature 125 \
-late_temperature 125 \
-corners {fast}

set_parasitics_parameters \
-early_spec tlup_min \
-late_spec tlup_min \
-early_temperature -40 \
-late_temperature -40 \
-corners {slow}

create_mode func
current_mode func
create_scenario -mode func -corner fast -name func_fast
create_scenario -mode func -corner slow -name func_slow

current_scenario func_slow
read_sdc $constraints_file

current_scenario func_fast
read_sdc $constraints_file

set_scenario_status func_slow -none -setup true -hold false -leakage_power true -dynamic_power true -max_transition true -max_capacitance true -min_capacitance false -active true
set_scenario_status func_fast -none -setup false -hold true -leakage_power true -dynamic_power false -max_transition true -max_capacitance false -min_capacitance true -active true
```

11:41 PM 8/15/2025


```
192.168.0.10 - Remote Desktop Connection
- layermap /home/tools/14nm/tech/star_rc/saed14nm_tf_tf_tluplus.map \
-name tlup_min

read_parasitic_tech \
-tlup /home/tools/14nm/tech/star_rc/saed14nm_1p0m_cmax.tluplus \
-layermap /home/tools/14nm/tech/star_rc/saed14nm_tf_tf_tluplus.map \
-name tlup_max

set_parasitics_parameters \
-early_spec tlup_max \
-late_spec tlup_max \
-early_temperature 125 \
-late_temperature 125 \
-corners (fast)

set_parasitics_parameters \
-early_spec tlup_min \
-late_spec tlup_min \
-early_temperature -40 \
-late_temperature -40 \
-corners (slow)

create_mode func
current_mode func
create_scenario -mode func -corner fast -name func_fast
create_scenario -mode func -corner slow -name func_slow

current_scenario func_slow
read_sdc $Constraints_file

current_scenario func_fast
read_sdc $Constraints_file

set_scenario_status func_slow -none -setup true -hold false -leakage_power true -dynamic_power true -max_transition true -max_capacitance true -min_capacitance false -active true
set_scenario_status func_fast -none -setup false -hold true -leakage_power true -dynamic_power false -max_transition true -max_capacitance false -min_capacitance true -active true

## To remove duplicate modes, corners, scenarios, and to improve runtime and capacity without loss of constraints :
remove_duplicate_timing_contexts

# Grouping paths ---
group_path -name MACRO2REG -from [get_cells -physical_context -filter design_type==macro] -to [all_registers]
group_path -name REG2MACRO -from [all_registers] -to [get_cells -physical_context -filter design_type==macro]
group_path -name INPUTS -from [all_inputs]
group_path -name OUTPUTS -to [all_outputs]
group_path -name REG2REG -from [all_registers] -to [all_registers]

55.1 B
scripts: tcsh - Konsole [No Name] - GVIM routing.tcl (~/.jbl_14nm/vpd/scripts) - GV common_setup.tcl (~/.jbl_14nm/vpd/scripts) data_preparation.tcl (~/.jbl_14nm/vpd/scripts) - GV mcmnm.tcl (~/.jbl_14nm/vpd/scripts) - GV 11:41 PM 8/15/2025
```

```
192.168.0.10 - Remote Desktop Connection
date
##### FLOORPLAN #####
#setting pin locations and direction 2=out,4=in,1=inout(if available)

# SAFETY CHECK
#
check_netlist
check_timing
report_design_mismatch -verbose

##### create shape of block and proceed with next steps #####

# PLACE PINS
#set_block_pin_constraints -self -allowed_layers {M3 M4} -sides 2
#place_pins -self

#setting pin locations and direction 2=out,4=in,1=inout(if available)

initialize_floorplan -core_utilization 0.5 -core_offset {2}

set_block_pin_constraints -self -allowed_layers {M3 M4} -sides 2
place_pins -ports [get_ports -filter direction==out]
set_block_pin_constraints -self -allowed_layers {M3 M4} -sides 4
place_pins -ports [get_ports -filter direction==in]
#Use below command only if INOUT port is available
#set_block_pin_constraints -self -allowed_layers {M3 M4} -sides 1
#place_pins -ports [get_ports -filter direction==inout]
#set_individual_pin_constraints -allowed_layers {3 4} -location {180 240} -pin_spacing_distance 0.222 -ports [get_ports -filter direction==in]
#place_pins -ports [get_ports -filter direction==in]
#set_individual_pin_constraints -allowed_layers {3 4} -location {154.514 0} -pin_spacing_distance 0.222 -ports [get_ports -filter direction==in]
# fix the ports
set_attribute [get_ports *] physical_status fixed
get_attribute [get_ports *] is_fixed

#create_placement -floorplan
#####Place Macros and Proceed with next steps #####

#fix the macros
set_attribute [get_cells -physical_context -filter design_type==macro] physical_status fixed
get_attribute [get_cells -physical_context -filter design_type==macro] is_fixed

#reset_placement

# Creating KEEPOUT MARGIN for all MACRO {left,Top, right, bottom} these value are with 20nm pls change according to 14nm
create_keepout_margin -type hard -outer {1 1.2 1 1.2} [get_cells -physical_context -filter design_type==macro]

#ADD END CAP cells
#get_lib_cells */SHFILL*.RVT
#set_boundary_cell_rules -left_boundary_cell saed32rvt_c/SHFILL3_RVT -right_boundary_cell saed32rvt_c/SHFILL3_RVT
#set_boundary_cell_rules -top_boundary_cells saed14slvt_ss0p72v125c/SAEDSLVT14_CAPT2 -bottom_boundary_cells saed14slvt_ss0p72v125c/SAEDSLVT14_CAPB2 -right_boundary_cell saed14rvt_tt0p6v125c/SAEDRVT14_CAPBIN13 -left_boundary_cell saed14rvt_tt0p6v125c/SAEDRVT14_CAPBTAP6 -prefix endcap
```


192.168.0.10 - Remote Desktop Connection

powerplan.tcl (-jbl_14nm/pd/scripts) - GVIM6

```
date
## create the PG nets
create_net -power VDD
create_net -ground VSS

## Making Logical Connections
connect_pg_net -net VDD [get_pins -hierarchical "**/*/*/*/*VDD*"]
connect_pg_net -net VSS [get_pins -hierarchical "**/*/*/*/*VSS*"]

## Setting up the attribute for TIE cells
set_attribute [get_lib_cells */*TIE*] dont_touch false
set_lib_cell_purpose -include optimization [get_lib_cells */*TIE*]

### creating PG Rails
create_pg_mesh_pattern P_top_two \
  -layers { \
    { {horizontal_layer: M0} {width: 0.12} {spacing: interleaving} {pitch: 2.40} {offset: 0.0} {trim : true} } \
    { {vertical_layer: M8} {width: 0.12} {spacing: interleaving} {pitch: 2.40} {offset: 0.0} {trim : true} } \
  }

set_pg_strategy S_default_vddvss \
  -core \
  -pattern { {name: P_top_two} {nets:{VSS VDD}} } \
  -extension { {{stop:design_boundary_and_generate_pin}} }

compile_pg -strategies {S_default_vddvss}

#### Creating Standard cell rails
create_pg_std_cell_conn_pattern std_pattern -layers {M1} -check_std_cell_drc true -mark_as_follow_pin false

set_pg_strategy rail_strat -core -pattern {{name:std_pattern}{nets:{VDD VSS}}}
compile_pg -strategies rail_strat

# create_pg_std_cell_conn_pattern std_rail_conn1 -rail_width 0.068 -layers M1

#set_pg_strategy std_rail_1 -pattern {{name : std_rail_conn1} {nets: "VDD VSS"}} -core

#compile_pg -strategies std_rail_1

#### Creation of Vias b/w rails and PG straps
create_pg_vias -nets {VDD VSS} -from_layers M1 -to_layers M8 -drc no_check

# Check physical Connectivity of PG nets
check_pg_connectivity

#Check for DRC errors in the design,
check_pg_drc

### saving block
save_block -as powerplan_done
```

11:43 PM
8/15/2025

```
192.168.0.10 - Remote Desktop Connection
# Making Logical Connections
connect_pg_net -net VDD [get_pins -hierarchical "**/*/*/*VDD*"]
connect_pg_net -net VSS [get_pins -hierarchical "**/*/*/*VSS*"]

## Setting up the attribute for TIE cells
set_attribute [get_lib_cells */*TIE*] dont_touch false
set_lib_cell_purpose -include optimization [get_lib_cells */*TIE*]

### Creating PG Rails
create_pg_mesh_pattern P_top_two \
  -layers { \
    { (horizontal_layer: M0) (width: 0.12) (spacing: interleaving) (pitch: 2.40) (offset: 0.0) (trim : true) } \
    { (vertical_layer: M0) (width: 0.12) (spacing: interleaving) (pitch: 2.40) (offset: 0.0) (trim : true) } \
  }

set_pg_strategy S_default_vddvss \
  -core \
  -pattern { (name: P_top_two) (nets:{VSS VDD}) } \
  -extension { {{stop:design_boundary_and_generate_pin}} }

compile_pg -strategies {S_default_vddvss}

### Creating Standard cell rails
create_pg_std_cell_conn_pattern std_pattern -layers {M1} -check_std_cell_drc true -mark_as_follow_pin false

set_pg_strategy rail_strat -core -pattern {{name:std_pattern}}(nets:{VDD VSS}}
compile_pg -strategies rail_strat
#
#create_pg_std_cell_conn_pattern std_rail_conn1 -rail_width 0.068 -layers M1

#set_pg_strategy std_rail_1 -pattern {{name : std_rail_conn1}} (nets: "VDD VSS") -core
#compile_pg -strategies std_rail_1

### Creation of Vias b/w rails and PG straps
create_pg_vias -nets {VDD VSS} -from_layers M1 -to_layers M0 -drc no_check

# Check physical Connectivity of PG nets
check_pg_connectivity

#Check for DRC errors in the design.
check_pg_drc

### saving block
save_block -as powerplan_done

#To remove Power routes if you have any issue while building powerplan use all below cmds.

#remove_pg_strategies -all
#remove_pg_patterns -all
#remove_pg_regions -all
#remove_pg_via_master_rules -all
```

4,22 3

scripts : bash - Konsole Gvim 11:43

192.168.0.10 - Remote Desktop Connection

placement.tcl (-jbi_14nm/pd/scripts) - GVIM7

```
date
#Add I/O BUFFERS
#get_lib_cells */SAEDRVT14_BUF_20
add_buffer [get_mets -of [get_ports *]] [get_lib_cells */SAEDRVT14_BUF_20]
negnet_placement [get_ports *]
set_attribute [get_cells eco_cel*] physical_status fixed
#*****
#ADD Placement Blockages
#*****

# Grouping paths ---
group_path -name MACRO2REG -from [get_cells -physical_context -filter design_type==macro] -to [all_registers]
group_path -name REG2MACRO -from [all_registers] -to [get_cells -physical_context -filter design_type==macro]
group_path -name INPUTS2REG -from [all_inputs] -to [all_registers]
group_path -name REG2OUTPUTS -from [all_registers] -to [all_outputs]
group_path -name REG2REG -from [all_registers] -to [all_registers]
group_path -name INPUTS2OUTPUTS -from [all_inputs] -to [all_outputs]

#Pre-PLACEMENT CHECK----->>>for errors
check_design -checks pre_placement_stage

#****Run below command only when "SCANDEF" is not in design*****
set_app_options -name place.coarse.continue_on_missing_scandef -value true
#report_app_options place.coarse*

##placement density setting
#Specify a maximum density that controls how densely the tool can place cells in uncongested areas during wire-length-driven placement
set_app_options -name place.coarse.max_density -value 0.6

#Specify a maximum utilization that controls how densely the tool can place cells in less congested areas that surround highly congested areas, so
#that the cells in the congested areas can be spread out to reduce the congestion
set_app_options -name place.coarse.congestion_driven_max_util -value 0.6

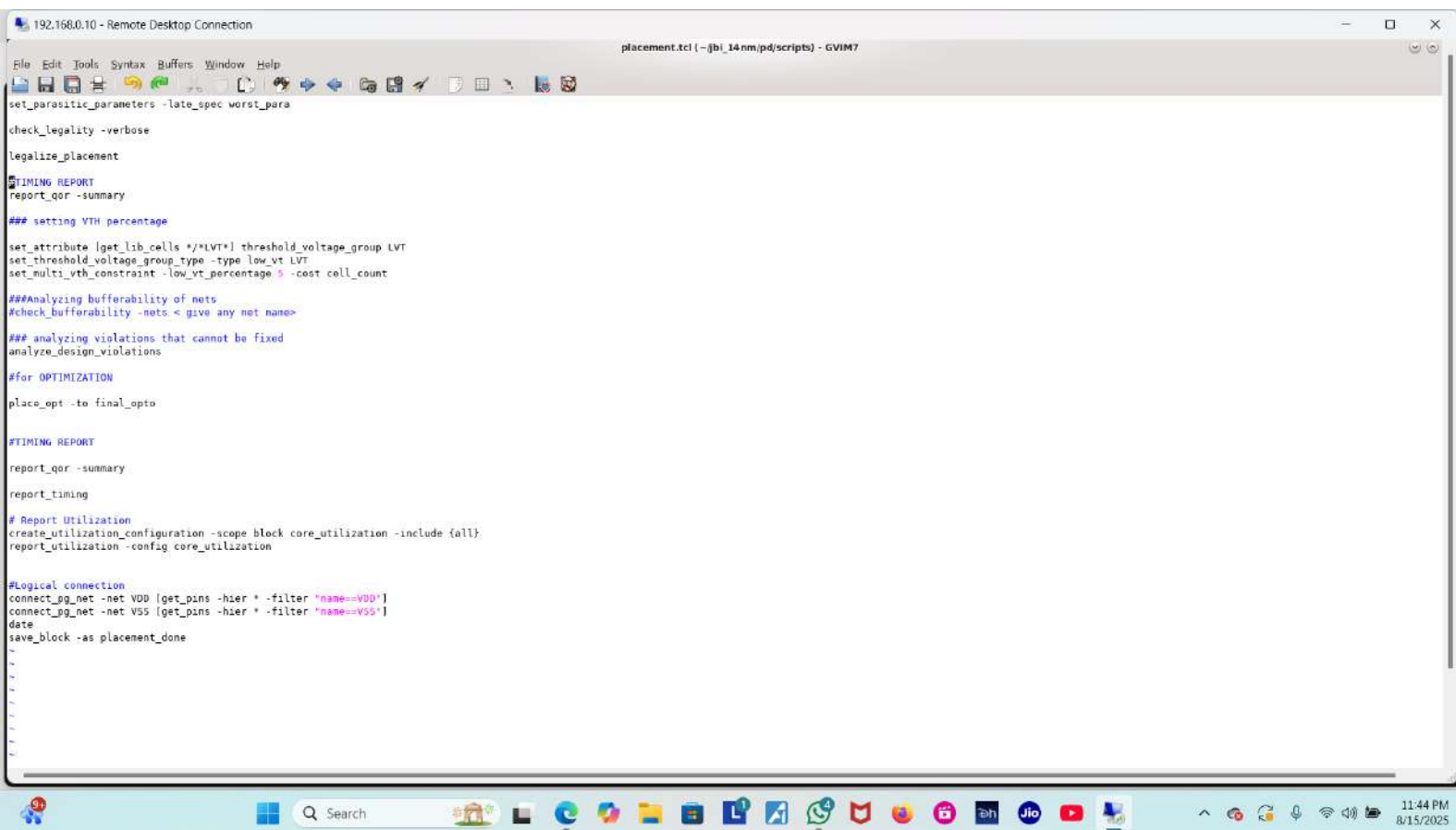
#to analyze all cells have proper or legal location
analyze_lib_cell_placement -lib_cells SAED*

#for placement of cells
create_placement -congestion

#set RC delay for timings which were set during 'DATA PREPARATION'
set_parasitic_parameters -early_spec best_para
set_parasitic_parameters -late_spec worst_para

check_legality -verbose
```

11:44 PM
8/15/2025



192.168.0.10 - Remote Desktop Connection

cts.tcl (-/jbl_14nm/pd/scripts) - GVIM8

```
date
set_clock_uncertainty -setup 0.0909 [get_clocks cmp_gclk]
set_clock_uncertainty -hold 0.1818 [get_clocks cmp_gclk]

set_clock_uncertainty -setup 0.1111 [get_clocks jbus_gclk]
set_clock_uncertainty -hold 0.2222 [get_clocks jbus_gclk]

update_timing -full

#Before performing CTS, execute the following command and analyze the report
check_design -checks pre_clock_tree_stage
# set HDR
#create_routing_rule clk_rule -widths {M6 0.112 M7 0.112 } -spacings {M6 0.112 M7 0.112 }
create_routing_rule clk_rule -widths {M6 0.18 M7 0.18 } -spacings {M6 0.08 M7 0.08 }
#check_clock_tree

# specify clock tree cell list
#set_lib_cell_purpose -exclude cts [get_lib_cells]
set_lib_cell_purpose -include cts [get_lib_cells "saed14rvt_tt0p6v125c/SAEDRVT14_BUF_5_2 saed14rvt_tt0p6v125c/SAEDRVT14_BUF_5_4 saed14rvt_tt0p6v125c/SAEDRVT14_BUF_5_6 saed14rvt_tt0p6v125c/SAEDRVT14_BUF_5_8"]

#Specify Max fanout
set_app_options -name cts.common.max_fanout -value 30

# set clock target skew and latency
set_clock_tree_options -clocks [all_clocks] -target_latency 0.4 -target_skew 0.05
#set_clock_tree_options -clocks [get_clocks -filter "is_virtual==false"] -target_latency 0.25 -target_skew 0.03

set_clock_routing_rules -clocks [all_clocks] -net_type {internal} -rules clk_rule -min_routing_layer M6 -max_routing_layer M7
set_clock_routing_rules -clocks [all_clocks] -net_type {root} -rules clk_rule -min_routing_layer M6 -max_routing_layer M7

clock_opt

# Make the logical connection of PG nets for all the standard cells
connect_pg_net -net VDD [get_pins -hier * -filter "name == VDD"]
connect_pg_net -net VSS [get_pins -hier * -filter "name == VSS"]

report_constraints -all_violators
report_clock_tree_options
report_clock
report_clock_qor
report_qor -summary
report_timing -delay_type min
report_timing -delay_type max
date
save_block -as cts_done
~
~
~
```

11:45 PM
8/15/2025

192.168.0.10 - Remote Desktop Connection

routing.tcl (-:/jbi_14nm/pd/scripts) - GVIM9

```
date
#check_design -checks pre_route_stage
check_routeability

# Run the following commands to set the options.

set_app_options -name route.detail.timing_driven -value true
set_app_options -name route.track.timing_driven -value true
set_app_options -name route.track.crosstalk_driven -value true
set_app_options -name route.global.timing_driven -value true

# Define the routing layers that needs to be used for routing
set_ignored_layers -max_routing_layer M7 -min_routing_layer M2

#Set the options to allow routing below the minimum layer only for pin connections.
set_app_options -name route.common.global_min_layer_mode -value allow_pin_connection

#Set the option to use a layer above the maximum layer as soft. This sets the tool to routing above the maximum layer is discouraged, but not disallowed.
set_app_options -name route.common.global_max_layer_mode -value soft

#Enabling the crosstalk aware and Delay calculations
set_app_options -name time.si_enable_analysis -value true
set_app_options -name time.enable_si_timing_windows -value true
set_app_options -name time.enable_ccs_rcv_cap -value true

#
set_attribute [get_layers M1] routing_direction vertical
create_via_def VIA12_Custon_short -cut_layer VIA1 -cut_size {0.03 0.03} -upper_layer M2 -lower_layer M1 -lower_enclosure {0.002 0.025} -upper_enclosure {0.025 0.002} -is_default
set_app_options -name route.common.connect_within_pins_by_layer_name -value {(M1 via_standard_cell_pins)}

#
set_attribute [get_layers M2] routing_direction horizontal
create_via_def VIA23_Custon_short -cut_layer VIA2 -cut_size {0.03 0.03} -upper_layer M3 -lower_layer M2 -lower_enclosure {0.002 0.025} -upper_enclosure {0.025 0.002} -is_default
set_app_options -name route.common.connect_within_pins_by_layer_name -value {(M2 via_standard_cell_pins)}

#route_global
#route_track
#route_detail
route_auto -max_detail_route_iterations 10

#route_auto
route_track -max_detail_route_iterations 10
```

11:45 PM 8/15/2025

```
192.168.0.10 - Remote Desktop Connection

set_app_options -name route.common.global_max_layer_mode -value soft

#Enabling the Crosstalk aware and Delay Calculations
set_app_options -name time.si_enable_analysis -value true
set_app_options -name time.enable_si_timing_windows -value true
set_app_options -name time.enable_ccs_rcv_cap -value true

#
set_attribute [get_layers M1] routing_direction vertical
create_via_def VIA23_Custom_short -cut_layer VIA1 -cut_size {0.03 0.03} -upper_layer M2 -lower_layer M1 -lower_enclosure {0.002 0.025} -upper_enclosure {0.025 0.002} -is_default
set_app_options -name route.common.connect_within_pins_by_layer_name -value {(M1 via_standard_cell_pins)}

#
set_attribute [get_layers M2] routing_direction horizontal
create_via_def VIA23_Custom_short -cut_layer VIA2 -cut_size {0.03 0.03} -upper_layer M3 -lower_layer M2 -lower_enclosure {0.002 0.025} -upper_enclosure {0.025 0.002} -is_default
set_app_options -name route.common.connect_within_pins_by_layer_name -value {(M2 via_standard_cell_pins)}

#route_global
#route_track
#route_detail
route_auto -max_detail_route_iterations 10

#route_auto
save_block -as initial_route_newlib
route_opt
save_block -as route_done_newlib

#### helps to reduce drcs
route_eco -open_net_driven true -reuse_existing_global_route true -utilize_dangling_wires true

# VT distribution
#report_threshold_voltage_group

# Report Utilization
#create_utilization_configuration -scope block core_utilization -include {all}
#report_utilization -config core_utilization
date

53,0-1 4
scripts : tcsh - Konsole  Gvim 11:45 PM 8/15/2025
```

```

192.168.0.10 - Remote Desktop Connection
pt.cworst.tcl (~/.jbi_14nm/pd/scripts) - GVIM11
File Edit Tools Syntax Buffers Window Help
[Icons]

set std_cell_delay_corner "s0p72vm40c"
set macro_cell_delay_corner "s0p72vm40c"

set DESIGN_REF_PATH "/home/tools/14nm/"
set DESIGN_REF_PATH1 "/home/tools/14nm/"
#provide path for routed netlist
set input_verilog ".../outputs/jbi_routed.v"

#provide path for sdc
set input_sdc ".../inputs/jbi.sdc"

#provide path for spef
set input_spef ".../scripts/starRC/jbi.cworst.spef"

# all paths are set here according to run tool in work directory,make sure to check paths before executing it.

set my_design "jbi"

set search_path * \
${DESIGN_REF_PATH}/stdcell_rvt/db_nldm \
${DESIGN_REF_PATH}/stdcell_hvt/db_nldm \
${DESIGN_REF_PATH}/stdcell_lvt/db_nldm \
${DESIGN_REF_PATH}/SABD14nm_EDK_SRAN_v_05072020/lib/sran_lp \
${DESIGN_REF_PATH}/SABD14nm_EDK_IO_v_06052010/SABD14_EDK/lib/io_std/db_nldm \

set link_path ** \
${DESIGN_REF_PATH1}/stdcell_rvt/db_nldm/saed14rvt_tt0p8v125c.db \
${DESIGN_REF_PATH1}/stdcell_rvt/db_nldm/saed14rvt_tt0p8v125c.db \
${DESIGN_REF_PATH1}/stdcell_rvt/db_nldm/saed14rvt_ff0p7v125c.db \
${DESIGN_REF_PATH1}/stdcell_rvt/db_nldm/saed14rvt_ff0p88v125c.db \
${DESIGN_REF_PATH1}/stdcell_rvt/db_nldm/saed14rvt_s0p6v125c.db \
${DESIGN_REF_PATH1}/stdcell_rvt/db_nldm/saed14rvt_s0p72v125c.db \
${DESIGN_REF_PATH1}/stdcell_rvt/db_nldm/saed14rvt_s0p6vm40c.db \
${DESIGN_REF_PATH1}/stdcell_rvt/db_nldm/saed14rvt_s0p72vm40c.db \
${DESIGN_REF_PATH1}/stdcell_rvt/db_nldm/saed14rvt_ff0p7vm40c.db \
${DESIGN_REF_PATH1}/stdcell_rvt/db_nldm/saed14rvt_ff0p88vm40c.db \
${DESIGN_REF_PATH1}/stdcell_rvt/db_nldm/saed14rvt_tt0p8vm40c.db \
${DESIGN_REF_PATH1}/stdcell_hvt/db_nldm/saed14hvt_s0p6v125c.db \
${DESIGN_REF_PATH1}/stdcell_hvt/db_nldm/saed14hvt_s0p72v125c.db \
${DESIGN_REF_PATH1}/stdcell_hvt/db_nldm/saed14hvt_s0p6vm40c.db \
${DESIGN_REF_PATH1}/stdcell_hvt/db_nldm/saed14hvt_ff0p7v125c.db \
${DESIGN_REF_PATH1}/stdcell_hvt/db_nldm/saed14hvt_tt0p8v125c.db \
${DESIGN_REF_PATH1}/stdcell_hvt/db_nldm/saed14hvt_tt0p8vm40c.db \
${DESIGN_REF_PATH1}/stdcell_hvt/db_nldm/saed14hvt_s0p6vm40c.db \
${DESIGN_REF_PATH1}/stdcell_hvt/db_nldm/saed14hvt_s0p72vm40c.db \
${DESIGN_REF_PATH1}/stdcell_lvt/db_nldm/saed14lvt_s0p6vm40c.db \

```

192.168.0.10 - Remote Desktop Connection

pt_cworst.tcl (-f/bl_14nm/pd/scripts) - GVIM11

```
File Edit Tools Syntax Buffers Window Help
[DESIGN_REF_PATH]/SAED14nm_EDK_IO_v_06052019/SAED14_EDK/lib/io_std/db_nldm/saed14io_yb_ffop080vm40c_1p08v.db \
$(DESIGN_REF_PATH)/SAED14nm_EDK_IO_v_06052019/SAED14_EDK/lib/io_std/db_nldm/saed14io_fc_ffop080v125c_1p08v.db \
$(DESIGN_REF_PATH)/SAED14nm_EDK_IO_v_06052019/SAED14_EDK/lib/io_std/db_nldm/saed14io_yb_ffop080v125c_1p08v.db \
$(DESIGN_REF_PATH)/SAED14nm_EDK_IO_v_06052019/SAED14_EDK/lib/io_std/db_nldm/saed14io_fc_ttop08v125c_1p08v.db \
$(DESIGN_REF_PATH)/SAED14nm_EDK_IO_v_06052019/SAED14_EDK/lib/io_std/db_nldm/saed14io_yb_ttop08v125c_1p08v.db \
$(DESIGN_REF_PATH)/SAED14nm_EDK_IO_v_06052019/SAED14_EDK/lib/io_std/db_nldm/saed14io_fc_ttop08vm40c_1p08v.db \
$(DESIGN_REF_PATH)/SAED14nm_EDK_IO_v_06052019/SAED14_EDK/lib/io_std/db_nldm/saed14io_yb_ttop08vm40c_1p08v.db \
$(DESIGN_REF_PATH)/SAED14nm_EDK_PLL_v_06052019/SAED14_EDK/lib/pll/logic_synth/saed14pll_ttop08v125c.db \
$(DESIGN_REF_PATH)/SAED14nm_EDK_PLL_v_06052019/SAED14_EDK/lib/pll/logic_synth/saed14pll_ttop08vm40c.db \
$(DESIGN_REF_PATH)/SAED14nm_EDK_PLL_v_06052019/SAED14_EDK/lib/pll/logic_synth/saed14pll_ssop72vm40c.db \
$(DESIGN_REF_PATH)/SAED14nm_EDK_PLL_v_06052019/SAED14_EDK/lib/pll/logic_synth/saed14pll_ffop080v125c.db \
$(DESIGN_REF_PATH)/SAED14nm_EDK_PLL_v_06052019/SAED14_EDK/lib/pll/logic_synth/saed14pll_ssop72v125c.db \

```

```
catch {sh mkdir db rpts}

read_verilog -f${input_verilog}*
set link_create_black_boxes false
current_design my_design
link_design my_design
read_sdc -f${input_sdc}*
read_parasitics -f${input_spef}* -keep_capacitive_coupling -format SPEF
update_timing -full
save_session ./db/${my_design}.session
```

11:47 PM 8/15/2025